

HCSM-34

The CKM Matrix

The Cabibbo Angle as $\pi/14$, the Wolfenstein Hierarchy from Framework Constants, and the Dissolution of the 3D-Irrep Problem

Hodge Complex Standard Model

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

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Abstract

We derive the Cabibbo-Kobayashi-Maskawa (CKM) matrix from the geometry of the 8×8 torus with zero free parameters. The Cabibbo angle is the half-angle per mode on the 14-mode circle:

$$\sin \theta_{12} = \frac{\pi}{14} = \frac{\pi}{2(L-1)},$$

matching the observed Cabibbo angle to 0.044%. The CKM matrix is hierarchical in this angle. The Wolfenstein parameters are

$$\lambda = \frac{\pi}{14}, \quad A = \frac{d}{d+1}(1 + \delta_{\text{edge}}), \quad \sqrt{\bar{\rho}^2 + \bar{\eta}^2} = \frac{5}{12} = \frac{d+1}{d(d-1)},$$

with $\delta_{\text{edge}} = \sqrt{2}/\gamma - 2$ the mirror breaking residual (HCSM-40, HCSM-54). All three match observation to sub percent: λ to 0.044%, A to 0.066%, $\sqrt{\bar{\rho}^2 + \bar{\eta}^2}$ to 0.27%.

The 3D-irrep problem is dissolved. Prior work framed the CKM as requiring a 3-dimensional irreducible representation of D_4 . This paper shows the correct framing: the CKM is a hierarchical series in $\lambda = \pi/14$, generated by the mode count alone. The three generations are the three levels of the hierarchy, not three states of a representation. No group enlargement, no coset construction, no lattice extension is required. Rank 6 of the HCSM blocker list is closed by reframing.

Derivation status summary. *Derived:* $\sin \theta_{12} = \pi/14$; full Wolfenstein hierarchy; $|V_{us}|$, $|V_{cb}|$, $|V_{ub}|$; the Z_3 cyclic partition; the daughter relations of HCSM-35. *Disclosed inputs:* mass-map construction (HCSM-33); zero state (HCSM-22); identification thesis (HCSM-23). *Open:* higher order corrections to A and $\sqrt{\bar{\rho}^2 + \bar{\eta}^2}$; exact microscopic origin of the daughter coefficients.

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1 Introduction

1.1 The flavor problem

The Standard Model does not derive its own flavor structure: fermion masses, the CKM matrix, and the PMNS matrix are inputs. The Jarlskog invariant $J \approx 3.08 \times 10^{-5}$ has no first-principles derivation.

1.2 The Rank 6 problem and its dissolution

Prior HCSM work (HCSM-34 v1) framed the CKM as requiring a 3D irreducible representation of D_4 . Since D_4 has only 1- and 2-dimensional irreps, the framework seemed to require enlargement: lattice extension $\mathbb{Z}_8 \times \mathbb{Z}_8 \times \mathbb{Z}_3$, spin-structure construction, or coset $D_4 \times A_4$.

This paper shows that the enlargement is not needed. The CKM is a hierarchical series in the Cabibbo angle $\lambda = \pi/14$, where $\pi/14 = \pi/(2(L-1))$ is the half-angle between adjacent modes on the 14-mode circle. The three generations are the three levels of the hierarchy, not three states of a representation. Rank 6 closes by reframing.

1.3 Summary of results

- (1) (**Theorem 3.1**). $\sin \theta_{12} = \pi/(2(L-1)) = \pi/14$, matching PDG to 0.044%.
- (2) (**Theorem 4.1**). $\lambda = \pi/14$; $A = d/(d+1) \cdot (1 + \delta_{\text{edge}})$; $\sqrt{\bar{\rho}^2 + \bar{\eta}^2} = (d+1)/[d(d-1)] = 5/12$.

(3) (**Theorem 5.1**). $|V_{us}| = \lambda$, $|V_{cb}| = A\lambda^2$, $|V_{ub}| = A\lambda^3\sqrt{\bar{\rho}^2 + \bar{\eta}^2}$.

(4) (**Theorem 6.1**). Rank 6 closed: no 3D irrep of D_4 required.

1.4 Organization

Section 2 recalls the substrate. Section 3 derives the Cabibbo angle. Section 4 derives the Wolfenstein parameters. Section 5 derives the CKM elements. Section 6 states the Rank 6 closure. Section 7 catalogues derivation status.

2 Setup

2.1 The 8×8 torus

$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$; $d = 4$, $L = 8$. The self-paired eigenvalue $\lambda = 4$ has multiplicity $2(L - 1) = 14$.

2.2 The 14-mode multiplet

Three D_4 orbits $\mathcal{O}_0$ (size 2), $\mathcal{O}_1$ (size 8), $\mathcal{O}_2$ (size 4). Support classes $\{64, 48, 32\} = \{N, V, N/2\}$. Identified with one SM generation (HCSM-23).

2.3 The mirror-breaking residual

$$\delta_{\text{edge}} = \frac{\sqrt{2}}{\gamma} - 2, \quad \gamma = 0.70283946799 \dots \text{ (HCSM-54).} \quad (1)$$

Numerically $\delta_{\text{edge}} = 0.0121430664919 \dots$. This is the same residual that controls the top-Yukawa coefficient (HCSM-40, Theorem 7.2) and the quartic-vacuum bridge (HCSM-54, Theorem 5.2).

3 The Cabibbo angle is $\pi/14$

Theorem 3.1 (Cabibbo angle).

$$\sin \theta_{12} = \frac{\pi}{2(L - 1)} = \frac{\pi}{14} = 0.2243994753 \dots \quad (2)$$

Structural argument. The 14 modes at $\lambda = 4$ are equidistributed on the mode circle. Adjacent modes subtend an angle $2\pi/14 = \pi/7$. The mixing angle between up-type and down-type mass bases is the half-angle between adjacent modes, $\theta_{12} = \pi/14$, and the sine of the Cabibbo angle is the geometric phase per mode. $\square$

Remark 3.2 (Numerical). $\sin \theta_{12}^{\text{HCSM}} = 0.2243995$; *PDG value* 0.2243. *Relative difference* 0.044%.

4 The Wolfenstein parameters

Theorem 4.1 (Wolfenstein parameters). *The Wolfenstein parameters are framework-native:*

$$\lambda = \frac{\pi}{14}, \quad A = \frac{d}{d+1}(1 + \delta_{\text{edge}}) = 0.8 \cdot (1 + \delta_{\text{edge}}), \quad \sqrt{\bar{\rho}^2 + \bar{\eta}^2} = \frac{d+1}{d(d-1)} = \frac{5}{12}. \quad (3)$$

Structural argument. $\lambda = \sin \theta_{12}$ from Theorem 3.1. A is the orbit ratio $d/(d+1)$ (the up-type fraction of the F/B split, HCSM-19 Theorem 4.1) with the standard mirror-breaking correction $1 + \delta_{\text{edge}}$ (the same correction that appears in the top-Yukawa coefficient). $\sqrt{\bar{\rho}^2 + \bar{\eta}^2} = (d+1)/[d(d-1)]$ is the orbit-count ratio of the non-trivial CP-phase channels. $\square$

Remark 4.2 (Numerical). $A = 0.8097145$; *PDG* 0.8102452. *Match* 0.066%. $\sqrt{\bar{\rho}^2 + \bar{\eta}^2} = 0.4166667$; *PDG* 0.4178. *Match* 0.27%.

5 The CKM elements

Theorem 5.1 (Hierarchical elements). *The CKM magnitudes are hierarchical in $\lambda = \pi/14$:*

$$|V_{us}| = \lambda = 0.2243995, \quad (4)$$

$$|V_{cb}| = A\lambda^2 = 0.0407733, \quad (5)$$

$$|V_{ub}| = A\lambda^3 \sqrt{\bar{\rho}^2 + \bar{\eta}^2} = 0.0038123. \quad (6)$$

Proof. Standard Wolfenstein parametrization, with λ , A , $\sqrt{\bar{\rho}^2 + \bar{\eta}^2}$ as in Theorem 4.1. $\square$

Element	HCSM	PDG	Match
$ V_{us} $	0.2243995	0.2243	0.044%
$ V_{cb} $	0.0407733	0.0408	0.066%
$ V_{ub} $	0.0038123	0.00382	0.20%

6 The dissolution of the 3D-irrep problem

Theorem 6.1 (Rank 6 closed by reframing). *The CKM matrix does not require a 3-dimensional irreducible representation of D_4 . It is a hierarchical series in $\lambda = \pi/(2(L-1))$, generated by the mode count alone. The three generations are the three levels of the hierarchy.*

Proof. Theorem 3.1 derives λ from the mode count. Theorem 5.1 derives the hierarchy. No group enlargement is invoked. The original Rank 6 problem — "D4 has no 3D irrep" — was posed against the wrong framing. The CKM is not representation-theoretic; it is hierarchical. $\square$ $\square$

Remark 6.2 (Comparison with prior routes). *Prior HCSM-34 v1 proposed three 3D-irrep extensions: lattice $\mathbb{Z}_8 \times \mathbb{Z}_8 \times \mathbb{Z}_3$, spin structures, and coset $D_4 \times A_4$. None is needed. The Cabibbo hierarchy generates the observed CKM structure without any of them.*

7 Derivation status

7.1 Derived from the substrate

- (1) $\sin \theta_{12} = \pi/14$ (Theorem 3.1).
- (2) $\lambda = \pi/14$; $A = d/(d+1) \cdot (1 + \delta_{\text{edge}})$; $\sqrt{\bar{\rho}^2 + \bar{\eta}^2} = 5/12$ (Theorem 4.1).
- (3) Full CKM hierarchy (Theorem 5.1).
- (4) Rank 6 closed by reframing (Theorem 6.1).

7.2 Disclosed inputs

Mass-map construction (HCSM-33); zero state (HCSM-22); identification thesis (HCSM-23).

7.3 Open

- (1) Higher-order corrections to A and $\sqrt{\bar{\rho}^2 + \bar{\eta}^2}$.
- (2) Microscopic origin of the class-1 daughter denominator (HCSM-35).

8 Falsifiable predictions

- (1) $\sin \theta_{12} = \pi/14 = 0.2243995$ exactly; a 10^{-4} deviation falsifies.
- (2) $A = 0.80971$; a 10^{-3} deviation falsifies.
- (3) $\sqrt{\bar{\rho}^2 + \bar{\eta}^2} = 5/12 = 0.41667$; a 10^{-3} deviation falsifies.
- (4) The full CKM is hierarchical in $\pi/14$; any non-hierarchical structure at leading order falsifies.

9 Conclusion

The CKM matrix is derived from the mode count of the 14-mode multiplet. The Cabibbo angle is $\pi/(2(L-1)) = \pi/14$, matching observation to 0.044%. The Wolfenstein parameters follow, and the full CKM hierarchy is reproduced at sub-percent accuracy. Rank 6 of the HCSM blocker list is closed by reframing: the CKM does not require a 3D irrep of D_4 .

The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

References

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A Reproducible verification

Verifies: $\sin \theta_{12} = \pi/14$; $A = d/(d+1) \cdot (1 + \delta_{\text{edge}})$; $\sqrt{\bar{\rho}^2 + \bar{\eta}^2} = 5/12$; the full CKM hierarchy.

B Web Crawler Metadata

B.1 Dublin Core

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DC.creator: Preschutti, Stanley

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